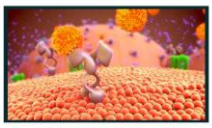
Chapter 2

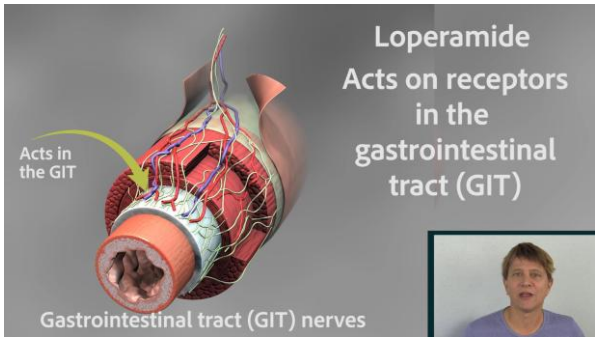
Introduction to medications that work on cellular receptors



Cellular receptors



1



2

Loperamide
(Imodium and others)

* An opioid
* Less than 1% is absorbed
* Works on the local cells

3

Safety issues concerning Diarrhoea (diarrhea)

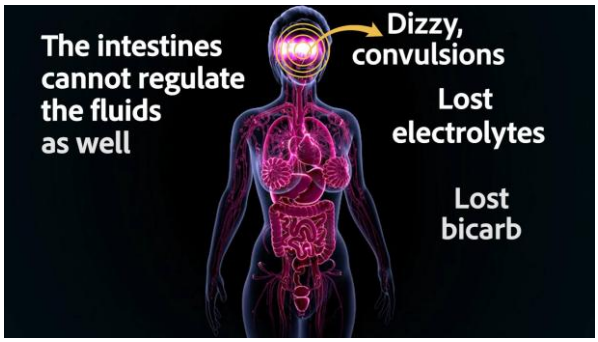
1. What are the main complications that can result from diarrhea?
2. What age groups and body shapes are most susceptible to the complications of diarrhea?
3. How would you know that a person is about to experience these complications?

4

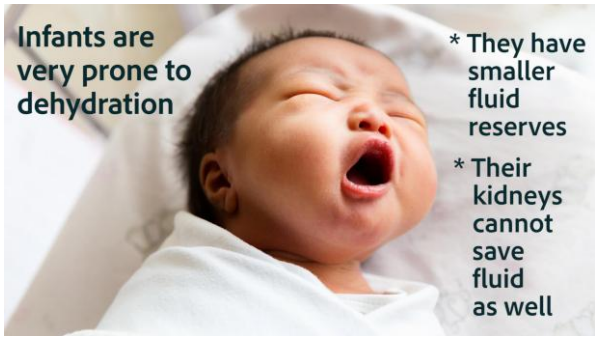
Safety issues concerning Diarrhoea (diarrhea)

1. What are the main complications that can result from diarrhea?
    * Dehydration
    * Electrolyte abnormalities
    * Acid-base abnormalities

5



6



7

Other people susceptible to dehydration

Obese individuals

* Adipose tissue only contains 10% fluid

The elderly

* Inadequate water intake,
* Excessive sweating,
* Medication like diuretics



8



9



10

Take the age of the individual into account when assessing skin turgor
(The elderly have naturally decreased turgor)



11

What is the primary therapy for diarrhea?

Fluids and electrolytes

Applying the facts, concepts and ideas

Understanding what the facts mean

Recognizing and recalling facts

12

Different names for loperamide

Loperamide

Lopedium

Imodium

Lomide

1-Piperidinebutanamide, 4-(4-chlorophenyl)-4-hydroxy-N,N-dimetyl-α,α-diphenyl-

Lopacut



13

What is the generic name?

The abbreviated version
of the chemical name
(it is assigned to the drug)

14

The generic name

A shortened version of the chemical name

Metoprolol

(R)-1-(isopropylamino)-3-
[4-(2-methoxyethyl)phenoxy]
propan-2-ol

15

All OLOL drugs are from the drug class termed beta blockers

Beta blockers block the adrenaline response at the level of the heart.



16

Prescription list

Propranolol

Prednisolone

Lorazepam

17

All OLOL drugs are from the drug class termed beta blockers

Propranolol



18

Always aim to spend time with each drug class that you learn, to fully picture it.

19

Beta blockers

Slow down heart; decrease blood pressure

Metoprolol
Propranolol
Atenolol
Nebivolol
Timolol

Look for the 'olol' suffix

5 of the top 200 most prescribed

20

Benzodiazepines

Anti anxiety; anti-seizure; CNS depressant

Alprazolam
Lorazepam
Clonazepam
Diazepam
Temazepam

'azepam' or 'azolam' suffix

5 of the top 200 most prescribed

21

Corticosteroids

Powerful immune system suppressants

Fluticasone
Hydrocortisone
Mometasone
Prednisolone
Methylprednisolone

'asone' ; 'pred ' ; 'cort' affixes (and others)

8 of the top 200 most prescribed

22

Opioids

Endogenous	Exogenous
Our natural opioids	Medications
Neurotransmitters	i.e. Loperamide

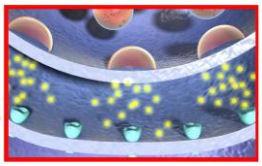


23

Opioids (some of the strongest pain killers)

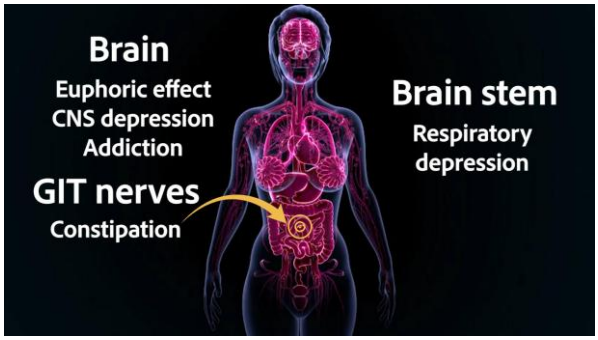
Morphine Codeine Heroin
Oxycodone Tramadol

Neuro-transmitters



Opioids are Inhibitory

24



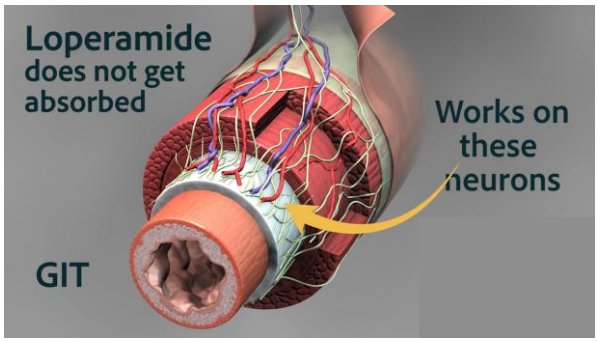
25

Loperamide

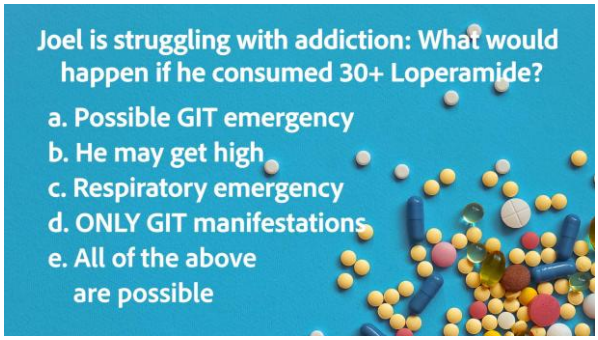
* **Indicated for the treatment of diarrhea**
* Not associated with side effects of opioids
  * No pain relief
  * No cognitive depression
  * No euphoria
  * No respiratory system depression



26



27



28

Loperamide is cardio toxic

Joel may die of an overdose if he consumes too much loperamide!

Over 100,000 people die each year from opioid abuse / misuse.

29

Only use medications in the way in which they are prescribed / recommended.

30

The phospholipid bilayer of cells

Proteins embedded in the cell membrane

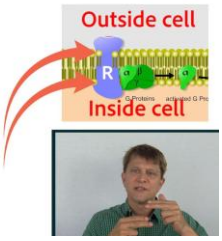
*Structural proteins
*Enzymes
*Channels
*Transport carriers
*Cellular receptors



31

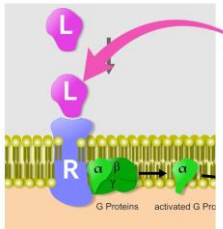
Cellular receptors

Ligand-gated ion channel
G-Protein linked receptor
Enzyme-linked receptor
Intracellular receptor



32

Cellular receptors

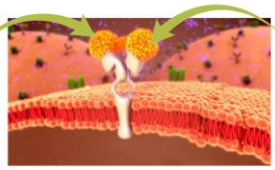


Ligand binds to receptor

33

Insulin receptor on cell surface

Ligand binding

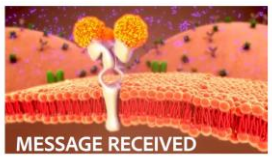


...an enzyme-linked receptor

34

Binding changes protein shape

Ligand binding

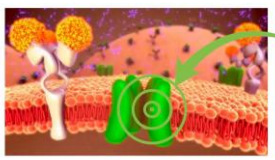


...an enzyme-linked receptor

35

Binding changes protein shape

Ligand binding

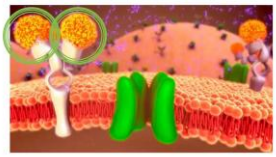


More Glut 4 carriers

...an enzyme-linked receptor

36

Ligands communicate with receptors



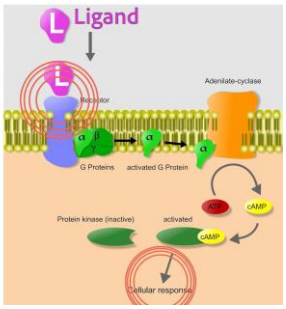
...an enzyme-linked receptor

37

G-Protein receptors

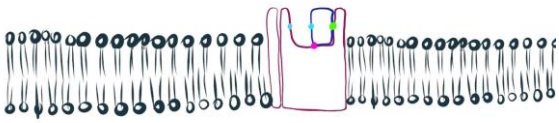
Also called 7-transmembrane receptors because they pass through the membrane 7 times

Largest group



38

Ligand gated ion channel



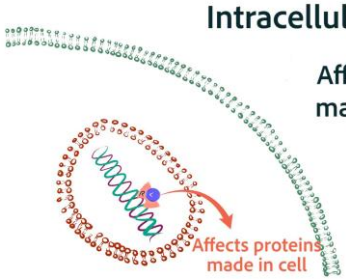
39

Ligand gated ion channel



40

Intracellular receptors



41

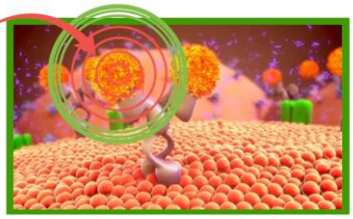
Most medications use cellular receptors to yield their therapeutic actions

42

Cellular receptors:
Most medications bind to one of them.

This binding

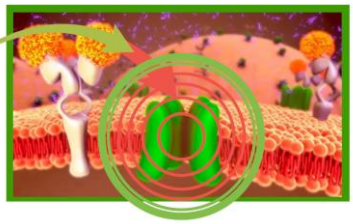
Results in a cellular response



43

Example: insulin binding to its receptors on muscle cells...

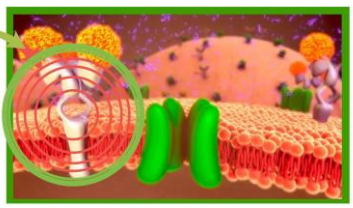
...results in more transport carriers



44

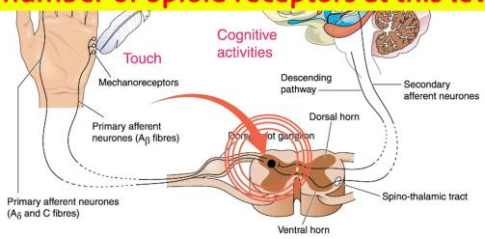
Our cells can also increase or decrease the number of receptors on its surface

Receptor number can be increased/ decreased



45

What would you predict will happen to the perception of pain when the cell decreases the number of opioid receptors at this level?



46

What would you predict will happen to the perception of pain when the cell decreases the number of opioid receptors at this level?

a. The perception of pain will stay the same

b. The perception of pain will increase

c. The perception of pain will decrease

47



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Understanding Medications

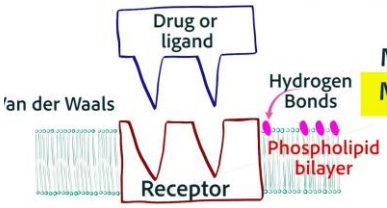
Chapter 2; Lesson E

*Receptor binding
*Receptor affinity
*Selectivity
*Potency



49

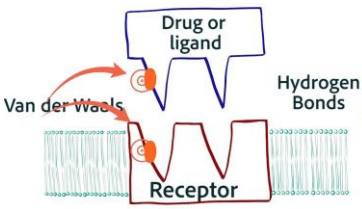
Receptor binding



Depends on:
Molecular shape
Molecular bonds

50

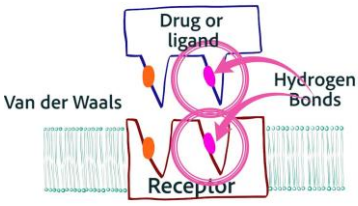
Receptor binding



Depends on:
Molecular shape
Molecular bonds

51

Receptor binding



Depends on:
Molecular shape
Molecular bonds



52

Forces involved in receptor binding

Strongest

Covalent
Ionic bonds
Hydrogen bonds
Hydrophobic bonds
Van der Waals

Weakest

53

Receptor binding

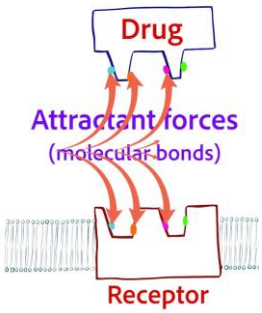


Attractant forces
(magnetic in this example)



54

Receptor binding



55

Affinity

Affinity is a measure of the strength of attraction between a receptor and its ligand

High affinity = strong attraction



56

Affinity

Affinity is a measure of the strength of attraction between a receptor and its ligand

Low affinity: less attraction; drug pops off receptor

57

Potency

Simplistically, how powerful the drug is

Drug A	Drug B
3mg dose	30mg dose

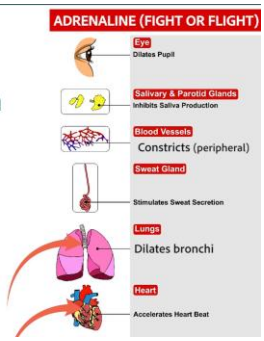
More potent

This may be because it has a greater affinity

58

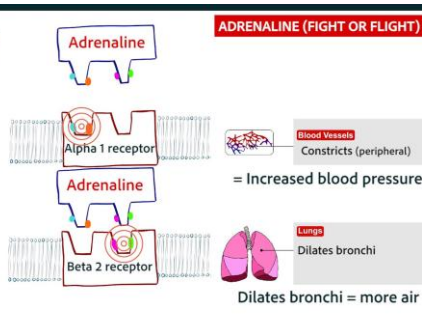
Selectivity

The degree to which a drug binds to a single receptor



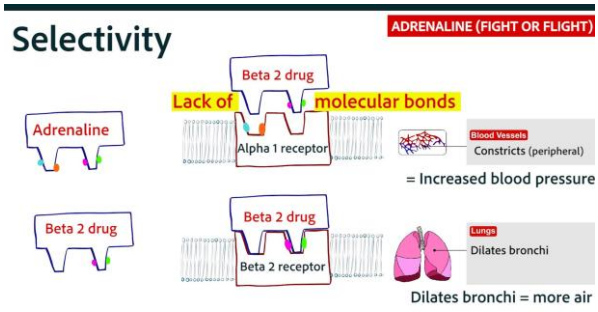
59

Selectivity

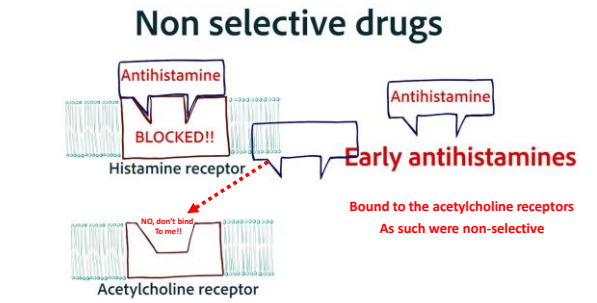


Adrenaline is non selective

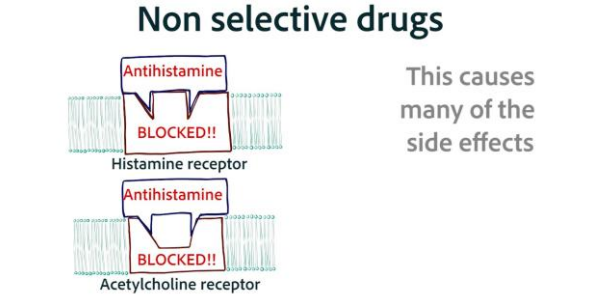
60


Selectivity
ADRENALINE (FIGHT OR FLIGHT)
Beta 2 drug
Lack of molecular bonds
Adrenaline
Alpha 1 receptor
Blood Vessels
Constricts (peripheral)
= Increased blood pressure
Beta 2 drug
Beta 2 drug
Beta 2 receptor
Lungs
Dilates bronchi
Dilates bronchi = more air

61


Non selective drugs
Antihistamine
BLOCKED!!
Histamine receptor
Antihistamine
Early antihistamines
Bound to the acetylcholine receptors
As such were non-selective
NO, don't bind To me!!
Acetylcholine receptor

62


Non selective drugs
Antihistamine
BLOCKED!!
Histamine receptor
Antihistamine
BLOCKED!!
Acetylcholine receptor
This causes many of the side effects

63

Summary

Affinity - How strong is the attraction?

Selectivity - How well does it bind to a single receptor?

Potency - How much do I take to get the effect?



64

Morphine
mu (μ) receptor

Fentynal
mu (μ) receptor
100 x more potent

Hypothesize on why?

Answer: The affinity to the receptor is much greater

65

Opioid mu (μ) receptor

Mu 1 receptor
Pain relief

Mu 2 receptor
Resp depression, constipation, addiction

Which is correct?

a. A selective mu 1 agonist would be valuable
b. An agonist with a strong affinlty to the mu 1 receptor would be potent
c. Both of the above

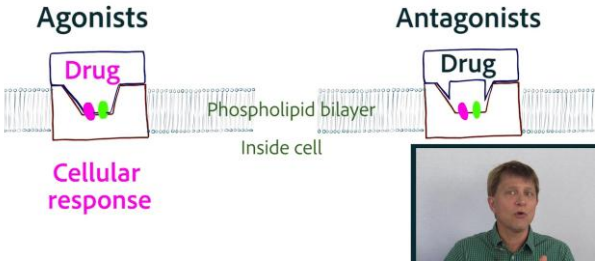
66

Terms used for medications that bind to cellular receptors

Affinity - how strong is the bond?
Selectivity - how well can it bind to one receptor without binding to others?
Potency - how much drug is needed to produce an effect?

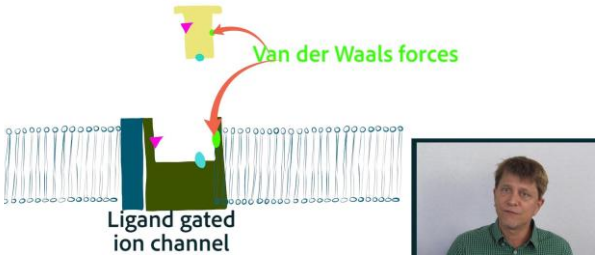
67

Cellular Receptors



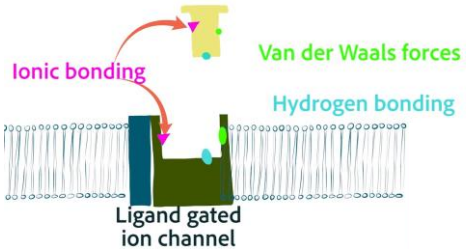
68

(Full) agonist



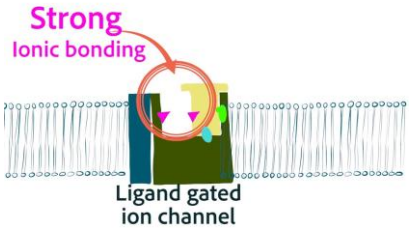
69

(Full) agonist



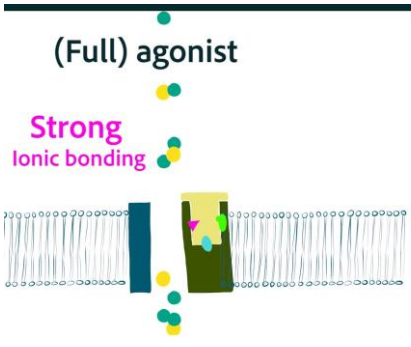
70

(Full) agonist



71

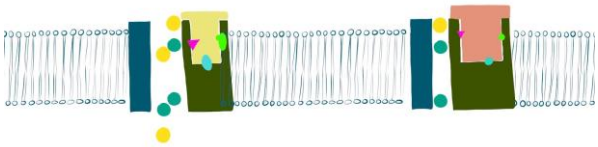
(Full) agonist



72

(Full) agonist
100% of activity

Partial agonist
Less than 100%



73

(Full) agonist

What would happen if the drug lacked the most important bond?

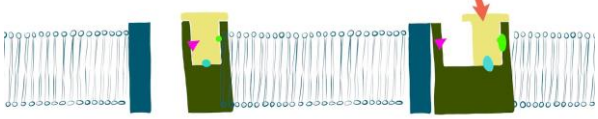


74

(Full) agonist

Antagonist (Competitive)

This drug is competing for the active spot



75

Inverse agonist

Normal receptor (functionally silent)

Receptor with constitutive activity

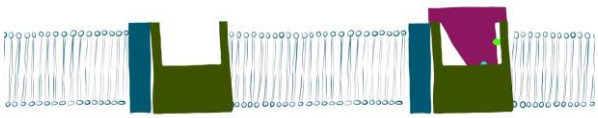


76

Inverse agonist

Normal receptor (functionally silent)

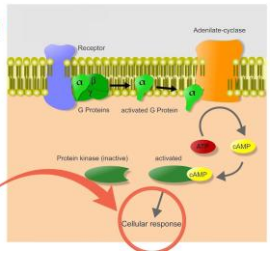
Receptor with constitutive activity



77

G-protein linked receptor

Unbound receptor



Still exhibits activity

Inverse agonist blocks the constitutive activity

78

Loperamide (an opioid)

Indicated in the treatment of diarrhea

Recall the main safety issues of diarrhea:

Fluid and electrolytes should be the first line treatment of diarrhea.

79

Other drugs that decrease diarrhea

* Most of the other opioids
  (These have notable side effects)
* Some of the anticholinergic agents (Hyoscyomine)
  (against the parasympathetic nervous system)
  * These are used mostly for diarrhea with spasms and not simple diarrhea

80

After rehydration and electrolyte therapy, loperamide is the preferred treatment for diarrhea.

81

Opioid antagonists

Methylnaltrexone

Is not absorbed / does not cross into the brain

Mainly used to counteract the constipating effects of opioid analgesics

Not to be used for general constipation

82

Opioid antagonists

Naloxone

Treats opioid overdose: saves many lives

VERY high affinity to opioid receptors

VERY short half life: needs to be re-administered

83

Chapter summary

* Safety first!
* Generic names are most important
* Opioid receptors are all through the body
* Loperamide works on cell receptors in the GIT



84

Coming up:
Pharmacokinetics

* Absorption
* Distribution
* Metabolism
* Excretion

85

References

- Dr Ken Wojcikowski
- PharmacoPhoto
- Royalty free images provided by Shutterstock

86